

CreditWise Loan Approval Prediction System

Project Report

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1. Project Overview

CreditWise is a machine learning-based loan approval prediction system designed to assist financial institutions in making faster and more consistent decisions. The project uses historical applicant data to predict whether a loan should be approved or rejected.

2. Problem Statement

Manual loan verification is time-consuming and prone to inconsistencies. The goal of this system is to reduce human effort while improving decision consistency and reducing financial risk.

3. Dataset

The project uses a structured dataset containing applicant demographics, financial information, loan details, and the target variable `Loan_Approved`. The dataset includes both numerical and categorical attributes.

4. Methodology

- Loaded and inspected the loan dataset using pandas.
- Handled missing values through imputation for both numerical and categorical columns.
- Performed exploratory data analysis using visualizations such as pie charts, histograms, and boxplots.
- Engineered new features such as squared DTI and Credit Score terms.
- Encoded categorical variables and scaled features for modeling.
- Trained multiple classifiers and compared their performance.

5. Models and Results

The project compared Logistic Regression, K-Nearest Neighbors, and Gaussian Naive Bayes. Logistic Regression produced the best overall performance with strong precision and recall balance.

Observed metrics: Accuracy 0.916, Precision 0.881, Recall 0.852, and F1-score 0.867.

6. Streamlit Deployment

The project was also deployed as a Streamlit web application so users can input applicant details and receive an instant loan approval prediction through the trained model(Logistic Regression).

7. Key Insights

Credit Score and Debt-to-Income Ratio were identified as more influential predictors than income alone. This highlights the importance of repayment behavior and financial discipline in loan approval decisions.

8. Conclusion

CreditWise demonstrates an end-to-end machine learning workflow for loan risk assessment. It combines data preprocessing, visualization, model training, evaluation, and deployment in a practical and user-friendly format.

Appendix

Files included in the project: `credit_wise.ipynb`, `loan_approval_data.csv`, `app.py`, `requirements.txt`, and this report.